

1.7 Introduction to Proofs

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"A proof is a valid argument which uses the hypothesis of the claim, previously verified theorems, facts, and axioms to establish truth of the claim."

Def: A theorem is a (usually very important) statement that can be shown to be true.

Def: A proposition is a (less significant) statement that can be shown to be true.

Def: An axiom is a statement assumed to be true.

Def: A lemma is a statement shown to be true, often used to prove larger theorems.

Def: A corollary is a result established from a theorem (special case).

Def: A conjecture is a claim that something is true based on evidence or intuition.

Axioms for Real Numbers (Not All)

- For x, y real numbers, $x+y$ is a real number.
- For x real number, $x+0=0+x=x$
- For x, y real numbers, either $x=y$ or $x>y$ or $x<y$.

Boils down to... the real numbers work how we expect.

To be formal, theorems should be stated

To be formal, theorems should be stated using quantifiers but this is not usually done in practice.

"For all positive real numbers x, y ,
if $x \cdot y > 1$, then $\frac{1}{x \cdot y} < 1$." } formal
 $\forall x, y \ P(x, y) \rightarrow Q(x, y).$

"If x, y are positive real numbers and
 $x \cdot y > 1$ then $\frac{1}{x \cdot y} < 1$." } common

How do we prove statements of this form?

To prove $\forall x, y \ (P(x, y) \rightarrow Q(x, y))$
we take arbitrary c, d such
that $P(c, d)$ and show $Q(c, d)$.

This way by universal generalization
since $P(c, d) \rightarrow Q(c, d)$ for arbitrary
 c, d , we have $\forall x, y \ (P(x, y) \rightarrow Q(x, y))$.

Def: A direct proof of a conditional
 $p \rightarrow q$ is made by assuming p is true
then through rules of inference, establish
 q is true.

Helpful properties of numbers

Def: An integer N is even iff there
is an integer k such that $N = 2 \cdot k$.

Def: An integer N is odd iff
there is an integer k such that
 $N = 2k + 1$.

Def: The parity of an integer is its
even-ness or odd-ness. Two integers
have the same parity if they are
both even or both odd.

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Use a direct proof to show

"If n is an even integer, then n^2 is an even integer."

$$\forall n (P(n) \rightarrow Q(n))$$

$$P(n) = "n \text{ is even.}" \quad Q(n) = "n^2 \text{ is even.}"$$

Let n be an arbitrary even integer. Then by the definition of even integers, there is an integer k such that $n = 2 \cdot k$. Squaring both sides yields $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$. Since $2 \cdot k^2$ is the product of integers, it is also an integer. We conclude n^2 is even since it is the product of 2 and an integer. Therefore if n is an even integer, n^2 is an even integer. $\square$

Use a direct proof to show...

"For perfect cubes m, n , the product $m \cdot n$ is also a perfect cube."

Let m and n be perfect cubes. Then there are integers s and t such that $m = s^3$ and $n = t^3$. Then

$$m \cdot n = s^3 \cdot t^3 = s \cdot s \cdot s \cdot t \cdot t \cdot t$$

$$= (st)(st)(st) = (st)^3. \text{ By the}$$

definition of perfect cube, we conclude $m \cdot n$ is a perfect cube. Therefore if m, n are perfect cubes, $m \cdot n$ is a perfect cube. $\square$